

De Gouden Tarwe SQL Database Challenge

(Day 5 – Part 2)

MySQL

Problem Statement

De Gouden Tarwe (Dutch for The Golden Wheat) is a premium bakery chain based in the Netherlands. Established in 2018, the company has grown rapidly and now operates more than 30 branches across 20 major Dutch cities including Amsterdam, Rotterdam, Utrecht, The Hague, Eindhoven, and Groningen.

The bakery sells authentic Dutch baked goods from everyday bread like tijgerbrood and volkorenbrood, to traditional pastries like stroopwafels, appeltaart, bossche bollen, and seasonal specialties like oliebollen (only sold in December) and pepernoten (only sold during the Sinterklaas season in November).

The company employs over 450 people across all branches: bakers, assistant bakers, sales staff, branch managers, delivery drivers, and support staff. It serves tens of thousands of customers, many of whom participate in a loyalty program with tiered benefits (Bronze, Silver, Gold, Platinum).

The Current Problem

De Gouden Tarwe currently stores all its operational data in flat CSV files. As the business has grown, this approach has become unmanageable:

1. Data redundancy: The same product name, price, and description are repeated in every transaction record.
2. No formal relationships: There is no way to enforce that a branch referenced in an order actually exists.
3. No data integrity: A typo in a city name creates a new city that does not actually exist.
4. Impossible to answer business questions: Simple queries like What is our best-selling product in Rotterdam? require manually cross-referencing multiple files.
5. Performance problems: With millions of rows, opening and searching CSV files is painfully slow.

Progress – Day 5

On Day 05, i will answer this question using SQL queries:

6. Which products have seasonal sales patterns?

To answer this question, we need the `summary_monthly_product_sales` table (a pre-aggregated summary table built from `orders`, `order_items`, and `products`, defined in `summary_table_monthly_sales.sql`) and the `products` table. The summary table provides monthly sales quantities (`total_qty`) per product, filtered by 'Voltooid' order status. We roll up the monthly data into quarterly totals, then compute the Coefficient of Variation ($CoV = \text{standard deviation} / \text{average}$) across the 4 quarters for each product to measure how unevenly sales are distributed throughout the year. The products table is used to retrieve the product name and `is_year_round` flag. A higher CoV indicates a stronger seasonal pattern — classified as 'Sterk Seizoensgebonden' ($CoV \geq 0.35$), 'Matig Seizoensgebonden' ($CoV \geq 0.20$), or 'Stabiel' ($CoV < 0.20$).

```

WITH quarterly_sales AS (
    SELECT
        s.product_id,
        CASE
            WHEN s.sale_month_num IN (1,2,3) THEN 1
            WHEN s.sale_month_num IN (4,5,6) THEN 2
            WHEN s.sale_month_num IN (7,8,9) THEN 3
            WHEN s.sale_month_num IN (10,11,12) THEN 4
        END AS quarter_num,
        SUM(s.total_qty) AS qty_sold,
        ROUND(SUM(s.total_revenue), 2) AS revenue
    FROM summary_monthly_product_sales s
    WHERE s.order_status = 'Voltooid'
    GROUP BY s.product_id,
        CASE
            WHEN s.sale_month_num IN (1,2,3) THEN 1
            WHEN s.sale_month_num IN (4,5,6) THEN 2
            WHEN s.sale_month_num IN (7,8,9) THEN 3
            WHEN s.sale_month_num IN (10,11,12) THEN 4
        END
),

product_stats AS (
    SELECT
        product_id,
        SUM(qty_sold) AS
total_qty,
        ROUND(AVG(qty_sold), 2) AS
avg_qty_per_quarter,
        ROUND(STDDEV_POP(qty_sold), 2) AS
stddev_qty,
        ROUND(STDDEV_POP(qty_sold) / NULLIF(AVG(qty_sold), 0), 4)
AS coeff_of_variation,
        MAX(qty_sold) AS
peak_quarter_qty
    FROM quarterly_sales
    GROUP BY product_id
),

peak_quarter AS (
    SELECT
        qs.product_id,
        CONCAT('Q', qs.quarter_num) AS peak_quarter_label,
        CASE qs.quarter_num
            WHEN 1 THEN 'Jan-Mar'
            WHEN 2 THEN 'Apr-Jun'
            WHEN 3 THEN 'Jul-Sep'
            WHEN 4 THEN 'Okt-Des'
        END AS peak_months,
        qs.qty_sold AS peak_qty
    FROM quarterly_sales qs
    JOIN product_stats ps
        ON qs.product_id = ps.product_id
        AND qs.qty_sold = ps.peak_quarter_qty
    -- resolve ties: keep the earliest quarter
    WHERE qs.quarter_num = (
        SELECT MIN(q2.quarter_num)
        FROM quarterly_sales q2
        WHERE q2.product_id = qs.product_id
        AND q2.qty_sold = ps.peak_quarter_qty
    )
)

```

```

)
-- Final result
SELECT
    p.product_name,
    IF(p.is_year_round, 'Ja', 'Nee') AS is_year_round,
    ps.total_qty,
    ps.avg_qty_per_quarter,
    ps.stddev_qty,
    ps.coeff_of_variation AS cov,
    pq.peak_quarter_label AS peak_quarter,
    pq.peak_months,
    pq.peak_qty,
    ROUND(pq.peak_qty / NULLIF(ps.total_qty, 0) * 100, 1) AS peak_share_pct,
    CASE
        WHEN ps.coeff_of_variation >= 0.35 THEN 'Sterk
Seizoensgebonden'
        WHEN ps.coeff_of_variation >= 0.20 THEN 'Matig
Seizoensgebonden'
        ELSE 'Stabiel'
    END AS seasonal_label
FROM product_stats ps
JOIN peak_quarter pq ON ps.product_id = pq.product_id
JOIN products p ON ps.product_id = p.product_id
ORDER BY ps.coeff_of_variation DESC, p.product_name;

```

Based on the query above, the results are:

Product name	Is year round	Total qty	Avg qty per quarter	Stddev qty	Cov	Peak quarter	Peak months	Peak qty	Peak share pct	Seasonal label
Oliebol Naturel	Nee	332743	166371,5	52683,5	0,3167	Q4	Okt-Des	219055	65,8	Matig Seizoensgebonden
Oliebol Rozijnen	Nee	333057	166528,5	52633,5	0,3161	Q4	Okt-Des	219162	65,8	Matig Seizoensgebonden
Kerststol	Nee	330686	165343	51647	0,3124	Q4	Okt-Des	216990	65,6	Matig Seizoensgebonden
Tompouce	Ja	1375673	343918,25	8746,08	0,0254	Q3	Jul-Sep	355118	25,8	Stabiel
Gevulde Koek	Ja	1372290	343072,5	8534,39	0,0249	Q3	Jul-Sep	353643	25,8	Stabiel
Bossche Bol	Ja	1374743	343685,75	8487,62	0,0247	Q3	Jul-Sep	355490	25,9	Stabiel
Afbakbroodje	Ja	1372087	343021,75	8445,89	0,0246	Q3	Jul-Sep	354565	25,8	Stabiel
Limburgse Vlaai Kers	Ja	1373984	343496	8453,48	0,0246	Q3	Jul-Sep	355154	25,8	Stabiel
Water (Spa Blauw)	Ja	1373826	343456,5	8463,96	0,0246	Q3	Jul-Sep	354925	25,8	Stabiel
Eierkoek	Ja	1372984	343246	8398,21	0,0245	Q3	Jul-Sep	354059	25,8	Stabiel
Roggebrood	Ja	1375083	343770,75	8393,77	0,0244	Q3	Jul-Sep	354773	25,8	Stabiel
Meergranen Brood	Ja	1370086	342521,5	8313,36	0,0243	Q3	Jul-Sep	351148	25,6	Stabiel

Product name	Is year round	Total qty	Avg qty per quarter	Stddev qty	Cov	Peak quarter	Peak months	Peak qty	Peak share pct	Seasonal label
Roomboterst aaf	Ja	1373088	343272	8213,41	0,0239	Q3	Jul-Sep	354178	25,8	Stabiel
Rozijnenbro od	Ja	1371808	342952	8174,86	0,0238	Q3	Jul-Sep	354011	25,8	Stabiel
Speltbrood	Ja	1374270	343567,5	8186,98	0,0238	Q3	Jul-Sep	353199	25,7	Stabiel
Koffie	Ja	1372863	343215,75	8052,21	0,0235	Q3	Jul-Sep	353470	25,7	Stabiel
Worstenbroo dje	Ja	1373968	343492	8067,43	0,0235	Q3	Jul-Sep	354071	25,8	Stabiel
Pompoenpitt en Brood	Ja	1370504	342626	8006,29	0,0234	Q3	Jul-Sep	353054	25,8	Stabiel
Koffiebroodj e	Ja	1373404	343351	7949,67	0,0232	Q3	Jul-Sep	353417	25,7	Stabiel
Pistolet	Ja	1372282	343070,5	7958,63	0,0232	Q3	Jul-Sep	353653	25,8	Stabiel
Thee	Ja	1368203	342050,75	7926,02	0,0232	Q3	Jul-Sep	351750	25,7	Stabiel
Frikandelbro odje	Ja	1370561	342640,25	7910,79	0,0231	Q3	Jul-Sep	353081	25,8	Stabiel
Arnhemse Meisjes	Ja	1374565	343641,25	7902,02	0,023	Q3	Jul-Sep	353803	25,7	Stabiel
Limburgse Vlaai Rijst	Ja	1373946	343486,5	7885,18	0,023	Q3	Jul-Sep	353633	25,7	Stabiel
Kletskop	Ja	1374106	343526,5	7862,15	0,0229	Q3	Jul-Sep	352649	25,7	Stabiel
Wit Casino Brood	Ja	1376244	344061	7878,95	0,0229	Q3	Jul-Sep	354081	25,7	Stabiel
Chocoladem elk	Ja	1372695	343173,75	7810,25	0,0228	Q3	Jul-Sep	354204	25,8	Stabiel
Croissant Chocolate	Ja	1373367	343341,75	7817,19	0,0228	Q3	Jul-Sep	352866	25,7	Stabiel
Kaiserbroodj e	Ja	1373256	343314	7782,34	0,0227	Q3	Jul-Sep	354535	25,8	Stabiel
Puddingbroo dje	Ja	1368634	342158,5	7727,8	0,0226	Q3	Jul-Sep	352067	25,7	Stabiel
Appeltaart (punt)	Ja	1374446	343611,5	7729,89	0,0225	Q3	Jul-Sep	353476	25,7	Stabiel
Appeltaart (heel)	Ja	1378523	344630,75	7668,94	0,0223	Q3	Jul-Sep	354070	25,7	Stabiel
Espresso	Ja	1372050	343012,5	7654,01	0,0223	Q3	Jul-Sep	351855	25,6	Stabiel
Bonbons Doos (16 stuks)	Ja	1373871	343467,75	7582,16	0,0221	Q3	Jul-Sep	353874	25,8	Stabiel
Ciabatta	Ja	1372885	343221,25	7552,54	0,022	Q3	Jul-Sep	353006	25,7	Stabiel
Waldkorn Brood	Ja	1370792	342698	7505,1	0,0219	Q3	Jul-Sep	353252	25,8	Stabiel
Mergpijpje	Ja	1372522	343130,5	7465,42	0,0218	Q3	Jul-Sep	352270	25,7	Stabiel
Appelflap	Ja	1374604	343651	7461	0,0217	Q3	Jul-Sep	353042	25,7	Stabiel
Fris (cola/fanta/si nas)	Ja	1376478	344119,5	7464,83	0,0217	Q3	Jul-Sep	354192	25,7	Stabiel
Petit Four Assortiment	Ja	1370593	342648,25	7422,28	0,0217	Q3	Jul-Sep	352935	25,8	Stabiel
Roombroodj e	Ja	1371349	342837,25	7455,89	0,0217	Q3	Jul-Sep	352681	25,7	Stabiel
Stroopwafel (vers)	Ja	1374771	343692,75	7449,26	0,0217	Q3	Jul-Sep	352708	25,7	Stabiel

Product name	Is year round	Total qty	Avg qty per quarter	Stddev qty	Cov	Peak quarter	Peak months	Peak qty	Peak share pct	Seasonal label
Volkoren Tarwebrood	Ja	1370670	342667,5	7426,33	0,0217	Q3	Jul-Sep	351983	25,7	Stabiel
Amandelstaaf	Ja	1372532	343133	7377,61	0,0215	Q3	Jul-Sep	352450	25,7	Stabiel
Bonbons Doos (9 stuks)	Ja	1373022	343255,5	7364,25	0,0215	Q3	Jul-Sep	352583	25,7	Stabiel
Croissant Amandel	Ja	1372664	343166	7371,02	0,0215	Q3	Jul-Sep	352344	25,7	Stabiel
Jan Hagel	Ja	1373747	343436,75	7389,02	0,0215	Q3	Jul-Sep	352204	25,6	Stabiel
Haagse Kakker	Ja	1369416	342354	7327,33	0,0214	Q3	Jul-Sep	351636	25,7	Stabiel
Speculaas	Ja	1372600	343150	7343,42	0,0214	Q3	Jul-Sep	352143	25,7	Stabiel
Verse Jus d'Orange	Ja	1370929	342732,25	7330,22	0,0214	Q3	Jul-Sep	351496	25,6	Stabiel
Volkoren Bol	Ja	1369749	342437,25	7313,07	0,0214	Q3	Jul-Sep	351270	25,6	Stabiel
Bruin Casino Brood	Ja	1369760	342440	7288	0,0213	Q3	Jul-Sep	351244	25,6	Stabiel
Moorkop	Ja	1375289	343822,25	7294,08	0,0212	Q3	Jul-Sep	353178	25,7	Stabiel
Slagroomtaart	Nee	1368291	342072,75	7249,02	0,0212	Q3	Jul-Sep	349543	25,5	Stabiel
Croissant Naturel	Ja	1372494	343123,5	7223,51	0,0211	Q3	Jul-Sep	353368	25,7	Stabiel
Gevulde Speculaas	Ja	1371678	342919,5	7242,23	0,0211	Q3	Jul-Sep	353318	25,8	Stabiel
Fries Suikerbrood	Ja	1373698	343424,5	7222,91	0,021	Q3	Jul-Sep	352462	25,7	Stabiel
Olijvenbrood	Ja	1373705	343426,25	7199,53	0,021	Q3	Jul-Sep	352869	25,7	Stabiel
Brabants Suikerbrood	Ja	1376453	344113,25	7180,34	0,0209	Q3	Jul-Sep	354110	25,7	Stabiel
Bruine Bol	Ja	1375243	343810,75	7183,49	0,0209	Q3	Jul-Sep	353128	25,7	Stabiel
Rozijnenbol met Spijs	Ja	1377450	344362,5	7199,56	0,0209	Q3	Jul-Sep	353456	25,7	Stabiel
Krentenbrood	Ja	1374565	343641,25	7017,2	0,0204	Q3	Jul-Sep	352380	25,6	Stabiel
Pain au Chocolat	Ja	1371557	342889,25	6974,09	0,0203	Q3	Jul-Sep	351802	25,6	Stabiel
Wit Puntje	Ja	1373394	343348,5	6972,64	0,0203	Q3	Jul-Sep	352354	25,7	Stabiel
Latte Macchiato	Ja	1373394	343348,5	6903,59	0,0201	Q3	Jul-Sep	350997	25,6	Stabiel
Limburgse Vlaai Abrikoos	Ja	1374573	343643,25	6894,66	0,0201	Q3	Jul-Sep	352673	25,7	Stabiel
Cappuccino	Ja	1374167	343541,75	6771,24	0,0197	Q3	Jul-Sep	352403	25,6	Stabiel
Verse Muntthee	Ja	1373257	343314,25	6719,8	0,0196	Q3	Jul-Sep	352305	25,7	Stabiel
Saucijzenbroodje	Ja	1371942	342985,5	6644,31	0,0194	Q3	Jul-Sep	350867	25,6	Stabiel
Banketstaaf	Ja	1369930	342482,5	6541,7	0,0191	Q3	Jul-Sep	351191	25,6	Stabiel
Maisbroodje	Ja	1370762	342690,5	6530,43	0,0191	Q3	Jul-Sep	351268	25,6	Stabiel

Product name	Is year round	Total qty	Avg qty per quarter	Stddev qty	Cov	Peak quarter	Peak months	Peak qty	Peak share pct	Seasonal label
Zuurdesembrood	Ja	1373047	343261,75	6445,02	0,0188	Q3	Jul-Sep	350467	25,5	Stabiel
Wit Tijgerbrood	Ja	1373518	343379,5	6296,42	0,0183	Q3	Jul-Sep	351682	25,6	Stabiel
Boterkoek	Ja	1375290	343822,5	6046,3	0,0176	Q3	Jul-Sep	351185	25,5	Stabiel
Kruidnoten (zakje)	Nee	238418	238418	0	0	Q4	Okt-Dec	238418	100	Stabiel
Letterbanket	Nee	236778	236778	0	0	Q4	Okt-Dec	236778	100	Stabiel
Oranjetompouce	Nee	114604	114604	0	0	Q2	Apr-Jun	114604	100	Stabiel
Pepernoten (zakje)	Nee	237514	237514	0	0	Q4	Okt-Dec	237514	100	Stabiel

Based on the output table, the conclusions are as follows:

- 1) Highly Seasonal (Appears only in one quarter: peak share pct = 100%)
 - Kruidnoten, Pepernoten, and Letterbanket exclusively sold in Q4 (Oct – Dec) during the Sinterklaas season.
 - Oranjetompouce exclusively sold in Q2 (April – June) during the Koningsdag (King's Day) season)
- 2) Moderately Seasonal (CoV > 0,30, peak share pct ~65%)
 - Oliebol Naturel and Oliebol Rozijnen peak sales occur in Q4 (Oct – Dec), though minor sales are still recorded in other quarters.
 - Kerststol sales peak in Q4 (Oct – Dec) during the Christmast and New Year Season, though minor sales are still recorded in other quarters.
- 3) Stable Throughout the Year (CoV below 0,3)

The remaining 71 products are stable performance throughout the year, with sales distributed evenly across all quarter

Recommendation: De Gouden Tarwe must ensure that ingredients stocks for these seasonal products are sufficiently available leading up to Q4 (Oct – Dec) and Q2 (April for Oranjetompouce) as demand can increase significantly during these periods.

Glossary of terms:

Is year round	Indicates whether the products is available throughout the entire year (Ja for Yes and Nee for No)
Total Qty	Total number of units sold over the full year
Avg Qty per quarter	Average sales volume calculated per quarter (total qty ÷ 4)
Stddev Qty (standard deviation)	To measure the fluctuation in sales volume
CoV (Coefficient of Variation)	Ratio showing the level of sales variability (stddev ÷ avg)
Peak share pct	The percentage of the peak quarter's sales contribution relative to the product's total annual sales.

7. What are the top 5 most wasted product per branch?

To answer this question, we need the waste_log, branches, and products tables. The waste_log table records wasted product quantities and their associated cost values per branch. The branches table is used to retrieve the branch name, and the products

table is used to retrieve the product name. We aggregate total wasted quantity and total cost lost per product per branch, then use ROW_NUMBER() partitioned by branch and ordered by cost lost descending to rank and select the top 5 most costly wasted products for each branch.

```
WITH waste_ranked AS (
    SELECT
        b.branch_name,
        p.product_name,
        SUM(w.quantity) AS total_wasted,
        SUM(w.cost_value) AS total_cost_lost,
        ROW_NUMBER() OVER (
            PARTITION BY w.branch_id
            ORDER BY SUM(w.cost_value) DESC
        ) AS ranked
    FROM waste_log w
    JOIN branches b ON w.branch_id = b.branch_id
    JOIN products p ON w.product_id = p.product_id
    GROUP BY w.branch_id, b.branch_name, p.product_name
)
SELECT branch_name, product_name, total_wasted,
       ROUND(total_cost_lost, 2) AS total_cost_lost
FROM waste_ranked
WHERE ranked <= 5
ORDER BY branch_name, ranked;
```

The query result are:

Branch name	Product name	Total wasted	Total cost lost
De Gouden Tarwe Almere 1	Bonbons Doos (16 stuks)	991	€ 6.763,38
De Gouden Tarwe Almere 1	Slagroomtaart	860	€ 4.966,20
De Gouden Tarwe Almere 1	Limburgse Vlaai Abrikoos	1067	€ 4.668,02
De Gouden Tarwe Almere 1	Bonbons Doos (9 stuks)	1063	€ 4.650,54
De Gouden Tarwe Almere 1	Appeltaart (heel)	890	€ 4.516,46
De Gouden Tarwe Almere 2	Bonbons Doos (16 stuks)	922	€ 6.292,42
De Gouden Tarwe Almere 2	Appeltaart (heel)	1074	€ 5.450,31
De Gouden Tarwe Almere 2	Slagroomtaart	888	€ 5.127,82
De Gouden Tarwe Almere 2	Bonbons Doos (9 stuks)	939	€ 4.108,05
De Gouden Tarwe Almere 2	Limburgse Vlaai Abrikoos	936	€ 4.094,96
De Gouden Tarwe Amersfoort 1	Bonbons Doos (16 stuks)	854	€ 5.828,33
De Gouden Tarwe Amersfoort 1	Slagroomtaart	986	€ 5.693,83
De Gouden Tarwe Amersfoort 1	Appeltaart (heel)	897	€ 4.552,06
De Gouden Tarwe Amersfoort 1	Limburgse Vlaai Abrikoos	883	€ 3.863,02
De Gouden Tarwe Amersfoort 1	Limburgse Vlaai Kers	851	€ 3.723,06
De Gouden Tarwe Amersfoort 2	Bonbons Doos (16 stuks)	982	€ 6.701,93
De Gouden Tarwe Amersfoort 2	Slagroomtaart	1011	€ 5.838,20
De Gouden Tarwe Amersfoort 2	Limburgse Vlaai Abrikoos	1048	€ 4.584,89
De Gouden Tarwe Amersfoort 2	Bonbons Doos (9 stuks)	1006	€ 4.401,08
De Gouden Tarwe Amersfoort 2	Limburgse Vlaai Kers	956	€ 4.182,42
De Gouden Tarwe Amsterdam 1	Bonbons Doos (16 stuks)	1042	€ 7.111,41
De Gouden Tarwe Amsterdam 1	Slagroomtaart	872	€ 5.035,53
De Gouden Tarwe Amsterdam 1	Appeltaart (heel)	892	€ 4.526,71
De Gouden Tarwe Amsterdam 1	Limburgse Vlaai Rijst	1071	€ 4.310,38
De Gouden Tarwe Amsterdam 1	Limburgse Vlaai Abrikoos	901	€ 3.941,77
De Gouden Tarwe Amsterdam 2	Bonbons Doos (16 stuks)	920	€ 6.278,73

Branch name	Product name	Total wasted	Total cost lost
De Gouden Tarwe Amsterdam 2	Slagroomtaart	1062	€ 6.132,68
De Gouden Tarwe Amsterdam 2	Appeltaart (heel)	917	€ 4.653,50
De Gouden Tarwe Amsterdam 2	Limburgse Vlaai Kers	1007	€ 4.405,51
De Gouden Tarwe Amsterdam 2	Limburgse Vlaai Abrikoos	994	€ 4.348,63
De Gouden Tarwe Amsterdam 3	Bonbons Doos (16 stuks)	906	€ 6.183,25
De Gouden Tarwe Amsterdam 3	Appeltaart (heel)	1002	€ 5.084,87
De Gouden Tarwe Amsterdam 3	Slagroomtaart	875	€ 5.052,86
De Gouden Tarwe Amsterdam 3	Limburgse Vlaai Rijst	960	€ 3.863,70
De Gouden Tarwe Amsterdam 3	Limburgse Vlaai Kers	857	€ 3.749,27
De Gouden Tarwe Arnhem 1	Bonbons Doos (16 stuks)	965	€ 6.585,87
De Gouden Tarwe Arnhem 1	Slagroomtaart	901	€ 5.202,99
De Gouden Tarwe Arnhem 1	Appeltaart (heel)	936	€ 4.749,98
De Gouden Tarwe Arnhem 1	Bonbons Doos (9 stuks)	1001	€ 4.379,33
De Gouden Tarwe Arnhem 1	Limburgse Vlaai Abrikoos	977	€ 4.274,26
De Gouden Tarwe Arnhem 2	Bonbons Doos (16 stuks)	1176	€ 8.025,90
De Gouden Tarwe Arnhem 2	Slagroomtaart	1137	€ 6.565,85
De Gouden Tarwe Arnhem 2	Appeltaart (heel)	849	€ 4.308,41
De Gouden Tarwe Arnhem 2	Limburgse Vlaai Abrikoos	944	€ 4.129,93
De Gouden Tarwe Arnhem 2	Limburgse Vlaai Rijst	1011	€ 4.068,94
De Gouden Tarwe Breda	Bonbons Doos (16 stuks)	968	€ 6.606,35
De Gouden Tarwe Breda	Slagroomtaart	913	€ 5.272,29
De Gouden Tarwe Breda	Limburgse Vlaai Kers	1022	€ 4.471,14
De Gouden Tarwe Breda	Limburgse Vlaai Rijst	1109	€ 4.463,40
De Gouden Tarwe Breda	Appeltaart (heel)	825	€ 4.186,67
De Gouden Tarwe Delft	Bonbons Doos (16 stuks)	1077	€ 7.350,28
De Gouden Tarwe Delft	Appeltaart (heel)	1103	€ 5.597,43
De Gouden Tarwe Delft	Slagroomtaart	852	€ 4.919,97
De Gouden Tarwe Delft	Bonbons Doos (9 stuks)	1061	€ 4.641,78
De Gouden Tarwe Delft	Limburgse Vlaai Kers	972	€ 4.252,41
De Gouden Tarwe Den Bosch	Bonbons Doos (16 stuks)	951	€ 6.490,36
De Gouden Tarwe Den Bosch	Slagroomtaart	900	€ 5.197,22
De Gouden Tarwe Den Bosch	Limburgse Vlaai Kers	1056	€ 4.619,92
De Gouden Tarwe Den Bosch	Bonbons Doos (9 stuks)	1002	€ 4.383,64
De Gouden Tarwe Den Bosch	Appeltaart (heel)	861	€ 4.369,36
De Gouden Tarwe Den Haag 1	Bonbons Doos (16 stuks)	845	€ 5.766,91
De Gouden Tarwe Den Haag 1	Appeltaart (heel)	1023	€ 5.191,44
De Gouden Tarwe Den Haag 1	Slagroomtaart	797	€ 4.602,40
De Gouden Tarwe Den Haag 1	Limburgse Vlaai Rijst	1024	€ 4.121,26
De Gouden Tarwe Den Haag 1	Bonbons Doos (9 stuks)	837	€ 3.661,81
De Gouden Tarwe Den Haag 2	Bonbons Doos (16 stuks)	925	€ 6.312,92
De Gouden Tarwe Den Haag 2	Slagroomtaart	924	€ 5.335,76
De Gouden Tarwe Den Haag 2	Appeltaart (heel)	970	€ 4.922,55
De Gouden Tarwe Den Haag 2	Limburgse Vlaai Abrikoos	915	€ 4.003,06
De Gouden Tarwe Den Haag 2	Bonbons Doos (9 stuks)	857	€ 3.749,30
De Gouden Tarwe Den Haag 3	Bonbons Doos (16 stuks)	893	€ 6.094,48
De Gouden Tarwe Den Haag 3	Slagroomtaart	878	€ 5.070,20
De Gouden Tarwe Den Haag 3	Appeltaart (heel)	888	€ 4.506,44

Branch name	Product name	Total wasted	Total cost lost
De Gouden Tarwe Den Haag 3	Limburgse Vlaai Kers	969	€ 4.239,31
De Gouden Tarwe Den Haag 3	Limburgse Vlaai Abrikoos	921	€ 4.029,26
De Gouden Tarwe Dordrecht	Bonbons Doos (16 stuks)	839	€ 5.725,95
De Gouden Tarwe Dordrecht	Appeltaart (heel)	1039	€ 5.272,77
De Gouden Tarwe Dordrecht	Slagroomtaart	877	€ 5.064,38
De Gouden Tarwe Dordrecht	Bonbons Doos (9 stuks)	955	€ 4.178,04
De Gouden Tarwe Dordrecht	Limburgse Vlaai Rijst	967	€ 3.891,85
De Gouden Tarwe Eindhoven 1	Bonbons Doos (16 stuks)	856	€ 5.841,97
De Gouden Tarwe Eindhoven 1	Slagroomtaart	983	€ 5.676,49
De Gouden Tarwe Eindhoven 1	Appeltaart (heel)	1011	€ 5.130,56
De Gouden Tarwe Eindhoven 1	Limburgse Vlaai Abrikoos	1062	€ 4.646,14
De Gouden Tarwe Eindhoven 1	Limburgse Vlaai Kers	885	€ 3.871,83
De Gouden Tarwe Eindhoven 2	Bonbons Doos (16 stuks)	883	€ 6.026,27
De Gouden Tarwe Eindhoven 2	Slagroomtaart	1013	€ 5.849,71
De Gouden Tarwe Eindhoven 2	Appeltaart (heel)	938	€ 4.760,07
De Gouden Tarwe Eindhoven 2	Limburgse Vlaai Kers	1027	€ 4.493,06
De Gouden Tarwe Eindhoven 2	Limburgse Vlaai Abrikoos	892	€ 3.902,45
De Gouden Tarwe Eindhoven 3	Bonbons Doos (16 stuks)	920	€ 6.278,79
De Gouden Tarwe Eindhoven 3	Appeltaart (heel)	1094	€ 5.551,81
De Gouden Tarwe Eindhoven 3	Limburgse Vlaai Kers	1056	€ 4.619,84
De Gouden Tarwe Eindhoven 3	Slagroomtaart	771	€ 4.452,21
De Gouden Tarwe Eindhoven 3	Bonbons Doos (9 stuks)	962	€ 4.208,68
De Gouden Tarwe Groningen	Bonbons Doos (16 stuks)	953	€ 6.503,99
De Gouden Tarwe Groningen	Slagroomtaart	907	€ 5.237,62
De Gouden Tarwe Groningen	Appeltaart (heel)	901	€ 4.572,40
De Gouden Tarwe Groningen	Limburgse Vlaai Abrikoos	1019	€ 4.458,05
De Gouden Tarwe Groningen	Limburgse Vlaai Kers	978	€ 4.278,67
De Gouden Tarwe Haarlem	Slagroomtaart	1098	€ 6.340,54
De Gouden Tarwe Haarlem	Bonbons Doos (16 stuks)	831	€ 5.671,38
De Gouden Tarwe Haarlem	Appeltaart (heel)	988	€ 5.013,87
De Gouden Tarwe Haarlem	Limburgse Vlaai Kers	1047	€ 4.580,54
De Gouden Tarwe Haarlem	Bonbons Doos (9 stuks)	990	€ 4.331,17
De Gouden Tarwe Leeuwarden	Bonbons Doos (16 stuks)	990	€ 6.756,51
De Gouden Tarwe Leeuwarden	Slagroomtaart	980	€ 5.659,20
De Gouden Tarwe Leeuwarden	Appeltaart (heel)	996	€ 5.054,48
De Gouden Tarwe Leeuwarden	Limburgse Vlaai Kers	979	€ 4.283,01
De Gouden Tarwe Leeuwarden	Limburgse Vlaai Abrikoos	883	€ 3.863,05
De Gouden Tarwe Leiden 1	Bonbons Doos (16 stuks)	965	€ 6.585,89
De Gouden Tarwe Leiden 1	Bonbons Doos (9 stuks)	1206	€ 5.276,15
De Gouden Tarwe Leiden 1	Slagroomtaart	899	€ 5.191,45
De Gouden Tarwe Leiden 1	Limburgse Vlaai Kers	1078	€ 4.716,13
De Gouden Tarwe Leiden 1	Appeltaart (heel)	894	€ 4.536,78
De Gouden Tarwe Leiden 2	Bonbons Doos (16 stuks)	953	€ 6.503,99
De Gouden Tarwe Leiden 2	Slagroomtaart	907	€ 5.237,65
De Gouden Tarwe Leiden 2	Appeltaart (heel)	989	€ 5.018,90
De Gouden Tarwe Leiden 2	Limburgse Vlaai Abrikoos	1114	€ 4.873,62
De Gouden Tarwe Leiden 2	Limburgse Vlaai Kers	882	€ 3.858,73

Branch name	Product name	Total wasted	Total cost lost
De Gouden Tarwe Maastricht	Bonbons Doos (16 stuks)	976	€ 6.661,03
De Gouden Tarwe Maastricht	Slagroomtaart	1059	€ 6.115,40
De Gouden Tarwe Maastricht	Appeltaart (heel)	999	€ 5.069,74
De Gouden Tarwe Maastricht	Bonbons Doos (9 stuks)	1106	€ 4.838,62
De Gouden Tarwe Maastricht	Limburgse Vlaai Abrikoos	928	€ 4.059,90
De Gouden Tarwe Nijmegen	Bonbons Doos (16 stuks)	930	€ 6.347,09
De Gouden Tarwe Nijmegen	Slagroomtaart	1042	€ 6.017,19
De Gouden Tarwe Nijmegen	Appeltaart (heel)	1027	€ 5.211,77
De Gouden Tarwe Nijmegen	Limburgse Vlaai Kers	1019	€ 4.458,02
De Gouden Tarwe Nijmegen	Bonbons Doos (9 stuks)	1002	€ 4.383,68
De Gouden Tarwe Rotterdam 1	Bonbons Doos (16 stuks)	995	€ 6.790,60
De Gouden Tarwe Rotterdam 1	Slagroomtaart	877	€ 5.064,38
De Gouden Tarwe Rotterdam 1	Limburgse Vlaai Kers	1000	€ 4.374,93
De Gouden Tarwe Rotterdam 1	Appeltaart (heel)	835	€ 4.237,37
De Gouden Tarwe Rotterdam 1	Limburgse Vlaai Rijst	1050	€ 4.225,98
De Gouden Tarwe Rotterdam 2	Bonbons Doos (16 stuks)	858	€ 5.855,60
De Gouden Tarwe Rotterdam 2	Appeltaart (heel)	950	€ 4.821,01
De Gouden Tarwe Rotterdam 2	Slagroomtaart	787	€ 4.544,65
De Gouden Tarwe Rotterdam 2	Bonbons Doos (9 stuks)	963	€ 4.213,00
De Gouden Tarwe Rotterdam 2	Limburgse Vlaai Kers	955	€ 4.178,10
De Gouden Tarwe Rotterdam 3	Slagroomtaart	1091	€ 6.300,18
De Gouden Tarwe Rotterdam 3	Bonbons Doos (16 stuks)	883	€ 6.026,29
De Gouden Tarwe Rotterdam 3	Appeltaart (heel)	1099	€ 5.577,14
De Gouden Tarwe Rotterdam 3	Bonbons Doos (9 stuks)	1025	€ 4.484,26
De Gouden Tarwe Rotterdam 3	Limburgse Vlaai Abrikoos	987	€ 4.318,08
De Gouden Tarwe Tilburg 1	Slagroomtaart	958	€ 5.532,11
De Gouden Tarwe Tilburg 1	Bonbons Doos (16 stuks)	786	€ 5.364,22
De Gouden Tarwe Tilburg 1	Bonbons Doos (9 stuks)	1115	€ 4.878,05
De Gouden Tarwe Tilburg 1	Appeltaart (heel)	877	€ 4.450,54
De Gouden Tarwe Tilburg 1	Limburgse Vlaai Abrikoos	945	€ 4.134,30
De Gouden Tarwe Tilburg 2	Bonbons Doos (16 stuks)	821	€ 5.603,08
De Gouden Tarwe Tilburg 2	Appeltaart (heel)	962	€ 4.881,89
De Gouden Tarwe Tilburg 2	Slagroomtaart	803	€ 4.637,06
De Gouden Tarwe Tilburg 2	Limburgse Vlaai Abrikoos	1035	€ 4.528,00
De Gouden Tarwe Tilburg 2	Bonbons Doos (9 stuks)	959	€ 4.195,51
De Gouden Tarwe Utrecht 1	Bonbons Doos (16 stuks)	1019	€ 6.954,38
De Gouden Tarwe Utrecht 1	Slagroomtaart	963	€ 5.561,06
De Gouden Tarwe Utrecht 1	Limburgse Vlaai Kers	1096	€ 4.794,89
De Gouden Tarwe Utrecht 1	Appeltaart (heel)	920	€ 4.668,75
De Gouden Tarwe Utrecht 1	Bonbons Doos (9 stuks)	1018	€ 4.453,64
De Gouden Tarwe Utrecht 2	Bonbons Doos (16 stuks)	1025	€ 6.995,40
De Gouden Tarwe Utrecht 2	Slagroomtaart	995	€ 5.745,83
De Gouden Tarwe Utrecht 2	Appeltaart (heel)	919	€ 4.663,72
De Gouden Tarwe Utrecht 2	Limburgse Vlaai Kers	990	€ 4.331,22
De Gouden Tarwe Utrecht 2	Limburgse Vlaai Abrikoos	986	€ 4.313,67
De Gouden Tarwe Utrecht 3	Bonbons Doos (16 stuks)	1000	€ 6.824,80
De Gouden Tarwe Utrecht 3	Slagroomtaart	881	€ 5.087,49

Branch name	Product name	Total wasted	Total cost lost
De Gouden Tarwe Utrecht 3	Limburgse Vlaai Abrikoos	1006	€ 4.401,13
De Gouden Tarwe Utrecht 3	Bonbons Doos (9 stuks)	955	€ 4.177,99
De Gouden Tarwe Utrecht 3	Appeltaart (heel)	783	€ 3.973,50
De Gouden Tarwe Zwolle	Bonbons Doos (16 stuks)	931	€ 6.353,86
De Gouden Tarwe Zwolle	Slagroomtaart	891	€ 5.145,24
De Gouden Tarwe Zwolle	Appeltaart (heel)	877	€ 4.450,53
De Gouden Tarwe Zwolle	Bonbons Doos (9 stuks)	1009	€ 4.414,27
De Gouden Tarwe Zwolle	Limburgse Vlaai Kers	936	€ 4.094,92

Based on the table above:

- 1) Bonbons Doos (16 stuks) is the most costly wasted product across nearly all branches. Always consistently ranking #1 with cost losses ranging from €5,364 to €8,025 per branch. The highest loss was recorded at De Gouden Tarwe Arnhem 2 (€8,025.90 from 1,176 units wasted).
- 2) Slagroomtaart and Appeltaart (heel) consistently appear in the top 5 at every branch, indicating that these products carry high waste risk, likely due to their short shelf life and high unit cost.
- 3) Limburgse Vlaai variants (Abrikoos, Kers, and Rijst) often appear in positions 3-5, showing that vlaai product generally have a higher chance of being wasted.
- 4) The waste pattern is relatively uniform across all branches, the same products dominate the top 5 everywhere. This indicates the issue is systemic (related to product characteristics such as shelf life and price).

Recommendation: waste reduction efforts should prioritize Bonbons Doos (16 stuks), Slagroomtaart, and Appeltaart as they contribute that largest financial losses.

8. What is the profit margin per product (selling price vs cost price)

To answer this question, we need the `summary_monthly_product_sales` table (pre-aggregated from `orders`, `order_items`, and `products` — defined in `summary_table_monthly_sales.sql`), the `products` table, and the `product_categories` table. The `summary` table provides total quantity sold and total revenue per product (filtered by 'Voltooid' status). The `products` table supplies the selling price (`unit_price`) and `cost_price` to calculate profit per unit and margin percentage. The `product_categories` table is used to retrieve the category name. We also compute total COGS (cost of goods sold) and gross profit for each product.

```
WITH product_sales AS (
    SELECT
        product_id,
        SUM(total_qty) AS total_sold,
        SUM(total_revenue) AS total_revenue
    FROM summary_monthly_product_sales
    WHERE order_status = 'Voltooid'
    GROUP BY product_id
)
SELECT
    p.product_name,
    pc.category_name,
    p.unit_price AS selling_price,
    p.cost_price,
    ROUND(p.unit_price - p.cost_price, 2) AS profit_per_unit,
    ROUND((p.unit_price - p.cost_price) / p.unit_price * 100, 1) AS
margin_pct,
```

```

        COALESCE(ps.total_sold, 0) AS total_sold,
        ROUND(COALESCE(ps.total_revenue, 0), 2) AS total_revenue,
        ROUND(COALESCE(ps.total_sold, 0) * p.cost_price, 2) AS
total_cogs,
        ROUND(COALESCE(ps.total_revenue, 0) - COALESCE(ps.total_sold, 0)
* p.cost_price, 2)
        AS gross_profit
FROM products p
JOIN product_categories pc ON p.category_id = pc.category_id
LEFT JOIN product_sales ps ON p.product_id = ps.product_id
ORDER BY gross_profit DESC;

```

The results are:

- 1) Bonbons Doos (16 stuks) is the most profitable product with €17,023,838.89 in gross profit, due to its high price (€19.50) and high sales volume (~1.37M units).
- 2) Slagroomtaart and Appeltaart (heel) follow as the 2nd and 3rd most profitable products with €14.3M and €12.7M gross profits.
- 3) All products share a nearly identical profit margin of ~65%, this means the bakery applies the same profit percentage to every product, whether it is cheap or expensive.
- 4) Seasonal products (Kerststol, Letterbanket, Oranjetompouce, Oliebol, Kruidnoten, Pepernoten) have significantly lower gross profit. Not because of lower margins, but due to limited sales periods resulting in much lower total quantities sold (114K–333K units vs. ~1.37M for year-round products).
- 5) The cheapest products (Afbakbroodje at €0.40, Wit Puntje at €0.45, Bruine Bol at €0.50) earn the least total profit because each unit only gives a small profit of €0.26–€0.33, even with high sales numbers.

Product name	Category name	Selling price	Cost price	Profit per unit	Margin pct	Total sold	Total revenue	Total cogs	Gross profit
Bonbons Doos (16 stuks)	Bonbons	19,5	6,82	12,68	65	1373871	€ 26.393.639,11	€ 9.369.800,22	€ 17.023.838,89
Slagroomtaart	Taarten	16,5	5,77	10,73	65	1368291	€ 22.238.526,83	€ 7.895.039,07	€ 14.343.487,76
Appeltaart (heel)	Taarten	14,5	5,07	9,43	65	1378523	€ 19.691.128,21	€ 6.989.111,61	€ 12.702.016,60
Limburgse Vlaai Abrikoos	Vlaaien	12,5	4,38	8,12	65	1374573	€ 16.922.545,95	€ 6.020.629,74	€ 10.901.916,21
Limburgse Vlaai Kers	Vlaaien	12,5	4,38	8,12	65	1373984	€ 16.915.649,76	€ 6.018.049,92	€ 10.897.599,84
Bonbons Doos (9 stuks)	Bonbons	12,5	4,38	8,12	65	1373022	€ 16.903.042,42	€ 6.013.836,36	€ 10.889.206,06
Limburgse Vlaai Rijst	Vlaaien	11,5	4,02	7,48	65	1373946	€ 15.562.999,51	€ 5.523.262,92	€ 10.039.736,59
Petit Four Assortiment	Petit Fours	8,5	2,97	5,53	65,1	1370593	€ 11.475.017,75	€ 4.070.661,21	€ 7.404.356,54
Banketstaaf	Banketbakkerij	5,8	2,03	3,77	65	1369930	€ 7.826.225,92	€ 2.780.957,90	€ 5.045.268,02
Zuurdesembrood	Specialiteiten	5,2	1,82	3,38	65	1373047	€ 7.033.811,75	€ 2.498.945,54	€ 4.534.866,21
Amandelstaaf	Banketbakkerij	5,2	1,82	3,38	65	1372532	€ 7.030.675,48	€ 2.498.008,24	€ 4.532.667,24
Speltbrood	Specialiteiten	4,9	1,72	3,18	64,9	1374270	€ 6.632.316,37	€ 2.363.744,40	€ 4.268.571,97
Roomboterstaaf	Banketbakkerij	4,8	1,68	3,12	65	1373088	€ 6.491.857,83	€ 2.306.787,84	€ 4.185.069,99
Fries Suikerbrood	Specialiteiten	4,75	1,66	3,09	65,1	1373698	€ 6.426.427,50	€ 2.280.338,68	€ 4.146.088,82
Olijvenbrood	Specialiteiten	4,6	1,61	2,99	65	1373705	€ 6.223.758,39	€ 2.211.665,05	€ 4.012.093,34
Brabants Suikerbrood	Specialiteiten	4,5	1,57	2,93	65,1	1376453	€ 6.100.984,21	€ 2.161.031,21	€ 3.939.953,00
Krentenbrood	Specialiteiten	4,2	1,47	2,73	65	1374565	€ 5.686.104,96	€ 2.020.610,55	€ 3.665.494,41
Rozijnenbrood	Specialiteiten	4,1	1,43	2,67	65,1	1371808	€ 5.540.056,65	€ 1.961.685,44	€ 3.578.371,21
Pompoenpitten Brood	Volkorenbrood	3,8	1,33	2,47	65	1370504	€ 5.129.454,16	€ 1.822.770,32	€ 3.306.683,84
Appeltaart (punt)	Gebakjes	3,5	1,22	2,28	65,1	1374446	€ 4.738.766,56	€ 1.676.824,12	€ 3.061.942,44
Latte Macchiato	Koffie	3,5	1,22	2,28	65,1	1373394	€ 4.735.392,57	€ 1.675.540,68	€ 3.059.851,89
Gevulde Speculaas	Traditioneel	3,5	1,22	2,28	65,1	1371678	€ 4.730.022,28	€ 1.673.447,16	€ 3.056.575,12
Verse Jus d'Orange	Frisdranken	3,5	1,22	2,28	65,1	1370929	€ 4.727.024,73	€ 1.672.533,38	€ 3.054.491,35
Waldkorn Brood	Bruinbrood	3,5	1,22	2,28	65,1	1370792	€ 4.725.013,69	€ 1.672.366,24	€ 3.052.647,45
Meergranen Brood	Bruinbrood	3,45	1,21	2,24	64,9	1370086	€ 4.655.126,01	€ 1.657.804,06	€ 2.997.321,95
Roggebrood	Bruinbrood	3,2	1,12	2,08	65	1375083	€ 4.334.194,72	€ 1.540.092,96	€ 2.794.101,76
Arnhemse Meisjes	Luxe Koekjes	3,2	1,12	2,08	65	1374565	€ 4.332.567,11	€ 1.539.512,80	€ 2.793.054,31
Cappuccino	Koffie	3,2	1,12	2,08	65	1374167	€ 4.331.661,40	€ 1.539.067,04	€ 2.792.594,36

Product name	Category name	Selling price	Cost price	Profit per unit	Margin pct	Total sold	Total revenue	Total cogs	Gross profit
Volkoren Tarwebrood	Volkorenbrood	3,1	1,08	2,02	65,2	1370670	€ 4.185.533,02	€ 1.480.323,60	€ 2.705.209,42
Verse Muntthee	Thee	3	1,05	1,95	65	1373257	€ 4.057.895,37	€ 1.441.919,85	€ 2.615.975,52
Chocolademelk	Frisdranken	3	1,05	1,95	65	1372695	€ 4.056.190,98	€ 1.441.329,75	€ 2.614.861,23
Moorkop	Gebakjes	2,85	1	1,85	64,9	1375289	€ 3.860.335,54	€ 1.375.289,00	€ 2.485.046,54
Wit Tijgerbrood	Witbrood	2,85	1	1,85	64,9	1373518	€ 3.855.770,26	€ 1.373.518,00	€ 2.482.252,26
Boterkoek	Traditioneel	2,8	0,98	1,82	65	1375290	€ 3.793.274,60	€ 1.347.784,20	€ 2.445.490,40
Bossche Bol	Gebakjes	2,75	0,96	1,79	65,1	1374743	€ 3.723.278,58	€ 1.319.753,28	€ 2.403.525,30
Bruin Casino Brood	Bruinbrood	2,7	0,94	1,76	65,2	1369760	€ 3.642.429,21	€ 1.287.574,40	€ 2.354.854,81
Wit Casino Brood	Witbrood	2,5	0,88	1,62	64,8	1376244	€ 3.388.783,32	€ 1.211.094,72	€ 2.177.688,60
Tompouce	Gebakjes	2,5	0,88	1,62	64,8	1375673	€ 3.387.629,74	€ 1.210.592,24	€ 2.177.037,50
Koffie	Koffie	2,5	0,88	1,62	64,8	1372863	€ 3.380.959,69	€ 1.208.119,44	€ 2.172.840,25
Worstenbroodje	Gebakjes	2,4	0,84	1,56	65	1373968	€ 3.248.214,05	€ 1.154.133,12	€ 2.094.080,93
Kletskop	Luxe Koekjes	2,4	0,84	1,56	65	1374106	€ 3.248.118,72	€ 1.154.249,04	€ 2.093.869,68
Appelflap	Gebakjes	2,3	0,8	1,5	65,2	1374604	€ 3.113.896,82	€ 1.099.683,20	€ 2.014.213,62
Thee	Thee	2,3	0,8	1,5	65,2	1368203	€ 3.099.368,92	€ 1.094.562,40	€ 2.004.806,52
Fris (cola/fanta/sinas)	Frisdranken	2,2	0,77	1,43	65	1376478	€ 2.982.817,68	€ 1.059.888,06	€ 1.922.929,62
Mergpijpje	Gebakjes	2,2	0,77	1,43	65	1372522	€ 2.974.014,69	€ 1.056.841,94	€ 1.917.172,75
Espresso	Koffie	2,2	0,77	1,43	65	1372050	€ 2.973.462,19	€ 1.056.478,50	€ 1.916.983,69
Rozijnenbol met Spijs	Gebakjes	2,1	0,73	1,37	65,2	1377450	€ 2.849.652,78	€ 1.005.538,50	€ 1.844.114,28
Saucijzenbroodje	Gebakjes	2,1	0,73	1,37	65,2	1371942	€ 2.837.853,03	€ 1.001.517,66	€ 1.836.335,37
Kerststol	Banketbakkerij	8,5	2,97	5,53	65,1	330686	€ 2.768.431,07	€ 982.137,42	€ 1.786.293,65
Frikandelbroodje	Gebakjes	2	0,7	1,3	65	1370561	€ 2.700.296,69	€ 959.392,70	€ 1.740.903,99
Roombroodje	Gebakjes	1,95	0,68	1,27	65,1	1371349	€ 2.634.194,83	€ 932.517,32	€ 1.701.677,51
Gevulde Koek	Gebakjes	1,9	0,66	1,24	65,3	1372290	€ 2.568.248,42	€ 905.711,40	€ 1.662.537,02
Croissant Amandel	Luxe Broodjes	1,85	0,65	1,2	64,9	1372664	€ 2.501.588,33	€ 892.231,60	€ 1.609.356,73
Puddingbroodje	Gebakjes	1,85	0,65	1,2	64,9	1368634	€ 2.494.270,85	€ 889.612,10	€ 1.604.658,75
Water (Spa Blauw)	Frisdranken	1,8	0,63	1,17	65	1373826	€ 2.435.632,19	€ 865.510,38	€ 1.570.121,81
Haagse Kakker	Luxe Koekjes	1,8	0,63	1,17	65	1369416	€ 2.428.122,62	€ 862.732,08	€ 1.565.390,54

Product name	Category name	Selling price	Cost price	Profit per unit	Margin pct	Total sold	Total revenue	Total cogs	Gross profit
Pain au Chocolat	Luxe Broodjes	1,75	0,61	1,14	65,1	1371557	€ 2.363.976,99	€ 836.649,77	€ 1.527.327,22
Croissant Chocolade	Luxe Broodjes	1,65	0,58	1,07	64,8	1373367	€ 2.232.560,87	€ 796.552,86	€ 1.436.008,01
Koffiebroodje	Gebakjes	1,6	0,56	1,04	65	1373404	€ 2.164.661,22	€ 769.106,24	€ 1.395.554,98
Stroopwafel (vers)	Traditioneel	1,5	0,52	0,98	65,3	1374771	€ 2.031.099,34	€ 714.880,92	€ 1.316.218,42
Eierkoek	Traditioneel	1,4	0,49	0,91	65	1372984	€ 1.893.323,19	€ 672.762,16	€ 1.220.561,03
Ciabatta	Luxe Broodjes	1,4	0,49	0,91	65	1372885	€ 1.892.872,45	€ 672.713,65	€ 1.220.158,80
Croissant Naturel	Luxe Broodjes	1,25	0,44	0,81	64,8	1372494	€ 1.690.005,01	€ 603.897,36	€ 1.086.107,65
Jan Hagel	Traditioneel	1,2	0,42	0,78	65	1373747	€ 1.623.906,73	€ 576.973,74	€ 1.046.932,99
Letterbanket	Banketbakkerij	6,5	2,27	4,23	65,1	236778	€ 1.515.748,86	€ 537.486,06	€ 978.262,80
Speculaas	Traditioneel	0,8	0,28	0,52	65	1372600	€ 1.081.521,79	€ 384.328,00	€ 697.193,79
Pistolet	Witte Broodjes	0,6	0,21	0,39	65	1372282	€ 811.050,48	€ 288.179,22	€ 522.871,26
Kaiserbroodje	Witte Broodjes	0,55	0,19	0,36	65,5	1373256	€ 743.899,76	€ 260.918,64	€ 482.981,12
Maisbroodje	Bruine Broodjes	0,55	0,19	0,36	65,5	1370762	€ 742.566,02	€ 260.444,78	€ 482.121,24
Volkoren Bol	Bruine Broodjes	0,55	0,19	0,36	65,5	1369749	€ 742.075,16	€ 260.252,31	€ 481.822,85
Bruine Bol	Bruine Broodjes	0,5	0,17	0,33	66	1375243	€ 677.463,42	€ 233.791,31	€ 443.672,11
Kruidnoten (zakje)	Seizoenskoekjes	2,8	0,98	1,82	65	238418	€ 657.518,16	€ 233.649,64	€ 423.868,52
Wit Puntje	Witte Broodjes	0,45	0,16	0,29	64,4	1373394	€ 608.832,52	€ 219.743,04	€ 389.089,48
Pepernoten (zakje)	Seizoenskoekjes	2,5	0,88	1,62	64,8	237514	€ 584.994,96	€ 209.012,32	€ 375.982,64
Afbakbroodje	Witte Broodjes	0,4	0,14	0,26	65	1372087	€ 540.642,56	€ 192.092,18	€ 348.550,38
Oliebol Rozijnen	Seizoensgebak	1,4	0,49	0,91	65	333057	€ 459.294,64	€ 163.197,93	€ 296.096,71
Oliebol Naturel	Seizoensgebak	1,2	0,42	0,78	65	332743	€ 393.306,05	€ 139.752,06	€ 253.553,99
Oranjetompouce	Seizoensgebak	2,85	1	1,85	64,9	114604	€ 321.657,93	€ 114.604,00	€ 207.053,93

9. What is the running total of loyalty points per customer over time?

To answer this question, we need the `loyalty_transactions` and `customers` tables. The `loyalty_transactions` table records each point transaction (earned, redeemed, bonus, or expired) with its date and point value. The `customers` table is used to retrieve the customer's name and loyalty tier. We use a window function (`SUM ... OVER`) with `ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW` to calculate the running total of loyalty points for each customer over time, ordered by transaction date.

```
SELECT
    c.customer_id,
    CONCAT(c.first_name, ' ', c.last_name) AS customer_name,
    c.loyalty_tier,
    lt.transaction_date,
    lt.transaction_type,
    lt.points,
    SUM(lt.points) OVER (
        PARTITION BY lt.customer_id
        ORDER BY lt.transaction_date, lt.loyalty_transaction_id
        ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW
    ) AS running_total
FROM loyalty_transactions lt
JOIN customers c ON lt.customer_id = c.customer_id
WHERE c.is_loyalty_member = 1
ORDER BY c.customer_id, lt.transaction_date;
```

The query result consists of 200.000 rows of data across seven columns: `customer_id`, `customer_name`, `loyalty_tier`, `transaction_date`, `transaction_type`, `points`, and `running_total`. Here are the summary from the results:

- 1) The loyalty program has four transaction types, `Verdiend` (earned), `Ingewisseld` (redeemed), `Bonus`, and `Verlopen` (expired). Showing a complete points lifecycle.
- 2) Many customers have negative running totals, which means they have redeemed or lost (expired) more points than they earned. This could indicate data issues with initial point balances not being captured, or customers redeeming points earned before the tracking period.
- 3) The data runs from 2019 to 2023, showing that customers in every tier (`Brons`, `Zilver`, `Goud`, `Platina`) have been active in the loyalty program for a long time.
- 4) When points expire (`Verlopen`), customers can lose a lot of points at once. For example, customer 23 (Noah de Vries) lost 943 points from one expiration, and customer 301 (Esther Jansen) reached -2,305 points total after several redemptions and expirations.
- 5) The running total shows how each customer's point balance changes over time. This is helpful for checking if the current points are correct and for finding any unusual activity.

10. What is each supplier's delivery reliability rate and average delivery time?

To answer this question, we need the `purchase_orders` and `suppliers` tables. The `purchase_orders` table records all procurement orders with their status (`Geleverd`, `Geannuleerd`, `Deels Geleverd`), order date, delivery date, and total amount. The `suppliers` table provides the supplier name and specialty. We aggregate by supplier to

calculate total orders, delivery counts by status, delivery rate percentage, average delivery days (for completed deliveries only), and total spend.

```
SELECT
    s.supplier_name,
    s.specialty,
    COUNT(*) AS total_orders,
    SUM(CASE WHEN po.status = 'Geleverd' THEN 1 ELSE 0 END) AS
delivered,
    SUM(CASE WHEN po.status = 'Geannuleerd' THEN 1 ELSE 0 END) AS
cancelled,
    SUM(CASE WHEN po.status = 'Deels Geleverd' THEN 1 ELSE 0 END) AS
partially_delivered,
    ROUND(
        SUM(CASE WHEN po.status = 'Geleverd' THEN 1 ELSE 0 END)
        / NULLIF(COUNT(*), 0) * 100, 1
    ) AS delivery_rate_pct,
    ROUND(AVG(
        CASE WHEN po.status = 'Geleverd'
            THEN DATEDIFF(po.delivery_date, po.order_date)
        END
    ), 1) AS avg_delivery_days,
    ROUND(SUM(po.total_amount), 2) AS total_spend
FROM purchase_orders po
JOIN suppliers s ON po.supplier_id = s.supplier_id
GROUP BY s.supplier_id, s.supplier_name, s.specialty
ORDER BY delivery_rate_pct DESC, avg_delivery_days ASC;
```

Based on the query above, the results will be displayed in the table below.

Supplier name	Specialty	Total orders	Delivered	Cancelled	Partially delivered	Delivery rate pct	Avg delivery days	Total spend
Koninklijke Zeelandia	Bakgrondstoffen	1501	1214	60	85	80,9	3,1	€ 2.544.204,65
Dawn Foods	Bakgrondstoffen	1511	1220	56	76	80,7	3	€ 2.508.896,45
Aviko	Aardappelproducten	1469	1184	90	66	80,6	3	€ 2.508.484,12
Orkla Food Ingredients	Ingredienten	1529	1230	72	72	80,4	3	€ 2.549.062,38
Bakels Nederland	Meel en Grondstoffen	1540	1236	82	90	80,3	3	€ 2.520.841,99
CSM Bakery Solutions	Bakgrondstoffen	1511	1207	68	83	79,9	3	€ 2.504.310,24
Douwe Egberts Prof	Koffie en Thee	1526	1220	78	69	79,9	3	€ 2.486.589,83
Unifine Food and Bake	Ingredienten	1539	1229	79	86	79,9	3	€ 2.516.562,30
Van Geloven	Snacks	1553	1236	80	72	79,6	3	€ 2.617.316,57
FrieslandCampina	Zuivel	1538	1216	67	82	79,1	2,9	€ 2.590.462,00
Koopmans	Meel	1506	1190	85	81	79	3	€ 2.520.598,25
Sonneveld Group	Broodverbeteraars	1450	1137	74	86	78,4	2,9	€ 2.320.449,14

Based on the table above:

- 1) All suppliers have a similar delivery rate, between 78.4% and 80.9%. This means their performance is steady but not very high.
- 2) Koninklijke Zeelandia has the best delivery rate (80.9%), with 1,214 out of 1,501 orders delivered successfully. This makes it the most reliable supplier.

- 3) The average delivery time is almost the same for all suppliers (2.9–3.1 days), which suggests they all follow similar delivery rules or agreements.
- 4) Sonneveld Group has the lowest delivery rate (78.4%) and the lowest total spending (€2.32M), while Van Geloven has the highest total spending (€2.62M).
- 5) The number of cancelled orders is quite low (56–90 out of ~1,500 orders per supplier, about 4–6%). This is acceptable, but there is still room to improve.

Recommendation: Since all suppliers perform at a similar level, the bakery should focus on reducing partial deliveries and cancellations instead of changing suppliers. Setting stricter delivery agreements or combining orders could help push the delivery rate above 80%.